

**Applies to: Firetube, Watertube, Flextube, Electric, and HRSG units**  
**Frequency of Use: During abnormal operation, alarms, system faults, startup issues, or performance drops**

## 1. Common Failure Signs & Symptoms

| Symptom                      | Possible Cause                                 | Immediate Action                          |
|------------------------------|------------------------------------------------|-------------------------------------------|
| Low steam pressure           | Fouled tubes, water carryover, faulty control  | Check pressuretrol, inspect water level   |
| Water level fluctuation      | Feedwater pump short-cycling, LWCO stuck       | Verify pump control & LWCO float          |
| Frequent burner cycling      | Improper modulating controls, oversized burner | Review modulation range & burner settings |
| Flame failure                | Dirty flame sensor, blocked fuel nozzle        | Clean sensor & inspect fuel atomizer      |
| Excessive stack temperature  | Soot buildup, poor combustion                  | Perform combustion test & soot removal    |
| Blowdown water too hot/cold  | Malfunctioning blowdown valve or timer         | Check schedule & thermal loss points      |
| Vibrations or strange noises | Scale buildup or low water hammering           | Check DA tank & chemical balance          |